HERA time-matching review

14 October 2026 | Three real flights | Calibration deferred

Decision: continue timing review before real-data training. The local code has been updated and tested, but the checked timing estimates do not establish the required 2.5-second synchronization accuracy. No verified training table or real-data model has been produced.

Completed work

Reviewed Annafy's local and Colab notebooks and Emma's CW-analysis code. Checked the six original files for June 1, 2025; October 4, 2025; and March 28, 2026. The original file hashes match before and after the review. Google Drive and GitHub content were not changed.

What the flight comparisons show

Flight	Logger wraps	Combined-window curve offsets*	Longest accepted GPS gap
2025-06-01	4	-17.2 to -14.0 seconds	0.74 minutes
2025-10-04	3	-41.8 to -40.2 seconds	46.19 minutes
2026-03-28	4	-21.9 to -8.1 seconds	0.48 minutes

*Offsets are diagnostic estimates to ADD to CW time, across logger channels and 5/10/20-second bins. Poor shape matches and boundary optima are excluded from this range. These are not approved corrections, uncertainty bounds or final training settings. Estimates vary substantially by flight section.

Feature changes based on the professor's reply

The real-data feature profile uses altitude, pressure, inside-box temperature, latitude and longitude; engineered variants add log pressure and flight derivatives/phase. Magnetic field and acceleration are excluded because calibration or units are unresolved. Inside-box temperature is not used to calculate atmospheric density. No outside-air APRS temperature was merged. The synthetic demonstration is unchanged.

October GPS: the 46.2-minute interval is between onboard GPS readings passing the basic validity screen. Intermittent APRS reception does not itself explain this onboard issue. The gap is preserved.

Why the clock mapping remains unverified

1. A sign error is reproducible in the supplied notebooks

The reviewed correlation function compares the reference curve with **np.roll(source, k)**. The notebook then applies the negative of that shift to source timestamps. With this convention the shift itself is the offset to add. A known +60-second source delay requires -60 seconds; the supplied mapping adds +60 seconds and leaves a +120-second mismatch. Positive, negative and zero-delay controls were checked. The corrected local helper recovers the known delays within 0.1 seconds in these fixtures.

2. High curve correlation does not establish clock accuracy

Annafy's files already have timer correction and filtering applied. Their fixed flight windows are specific to that example. The new review uses sensor-selected windows for each of our flights and real overlapping samples rather than circular wrapping. The selected offsets still change with flight section and radiation channel. The same-clock logger-channel controls also show apparent shifts, despite sharing one timer. Pressure-based comparisons use no radiation counts, but their results also depend on pressure response and flight section.

3. What Emma's code establishes, and what was deferred

Emma's analysis code uses 4294.967296 seconds per timer wrap, corroborating the earlier wrap hypothesis. It is not the exact acquisition firmware. Her calibration workflow fits per-scintillator amplitude peaks in ground/coincidence data and can apply values from the same detector. No example calibration constants, particle labels or temperature-gain corrections were applied to these flights.

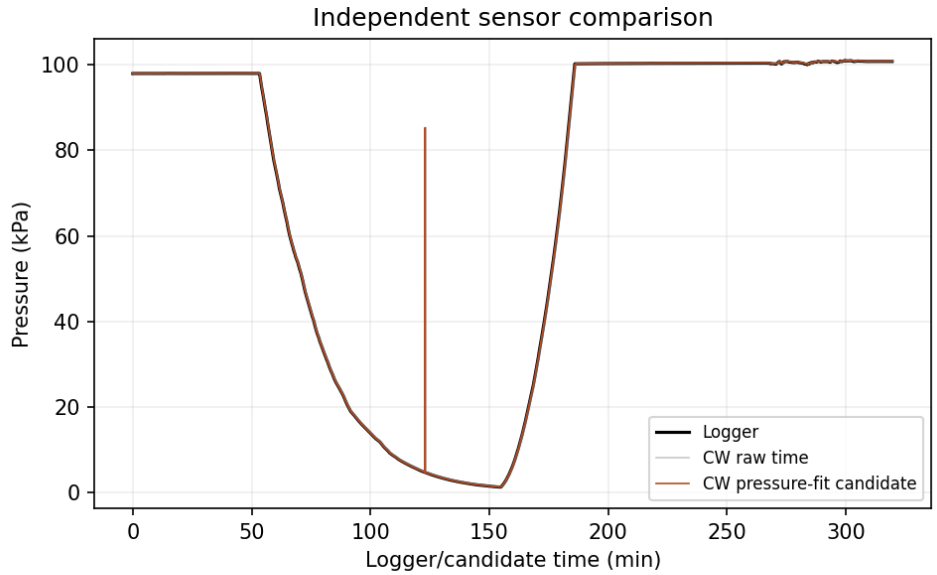
Next information to review with the team

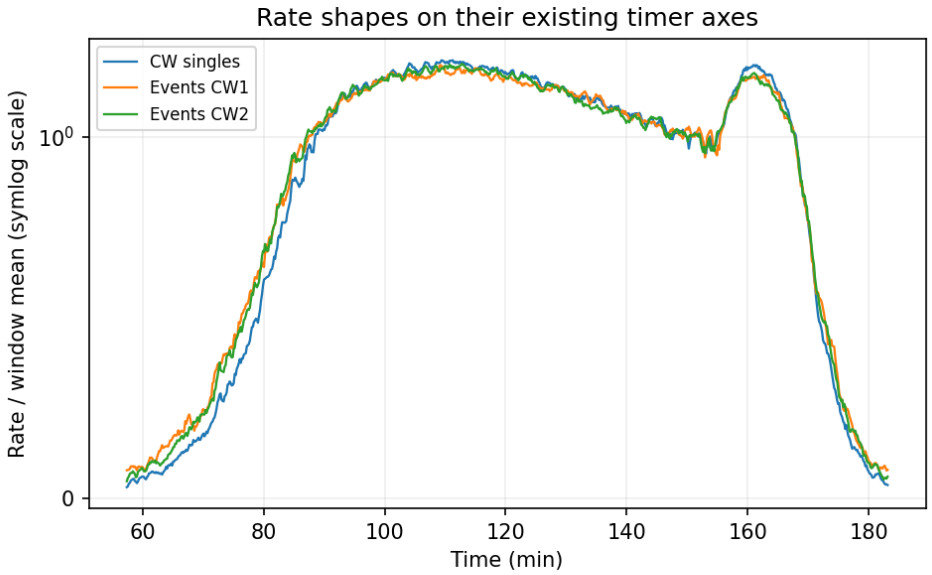
Review the sign-test result and establish a timing map, uncertainty and usable intervals for each exact file pair. Useful evidence includes shared events, sync pulses, start/end references, or the preparation records for Annafy's timer-corrected AHD011 file. Recording continuity/exposure and detector identity/mode still need confirmation, separately from the deferred amplitude calibration. Keep RUN_TRAINING=False.

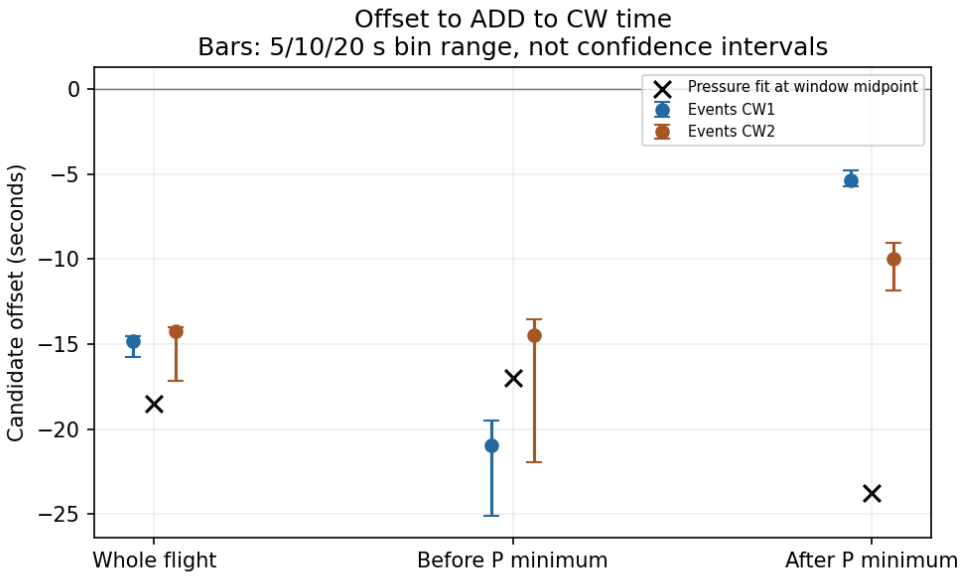
Validation: 35 project tests pass with XGBoost CPU 3.2.0. Model tests use artificial fixtures only. Full details, exact hashes, comparison CSVs and conditional review bins are in the add-on. Every review-bin row has training_ready=False; the bins are not combined.csv.

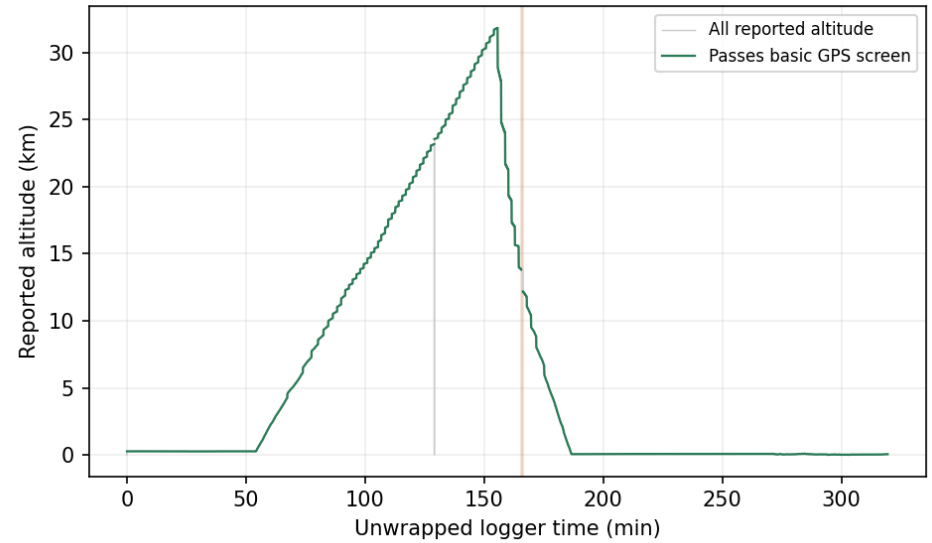
Sources: Annafy's shared folder; Emma's CW-analysis, commit 666ec5f; professor reply supplied by the user on 14 October 2026.

2025-06-01 | Timing comparison — diagnostic candidates only



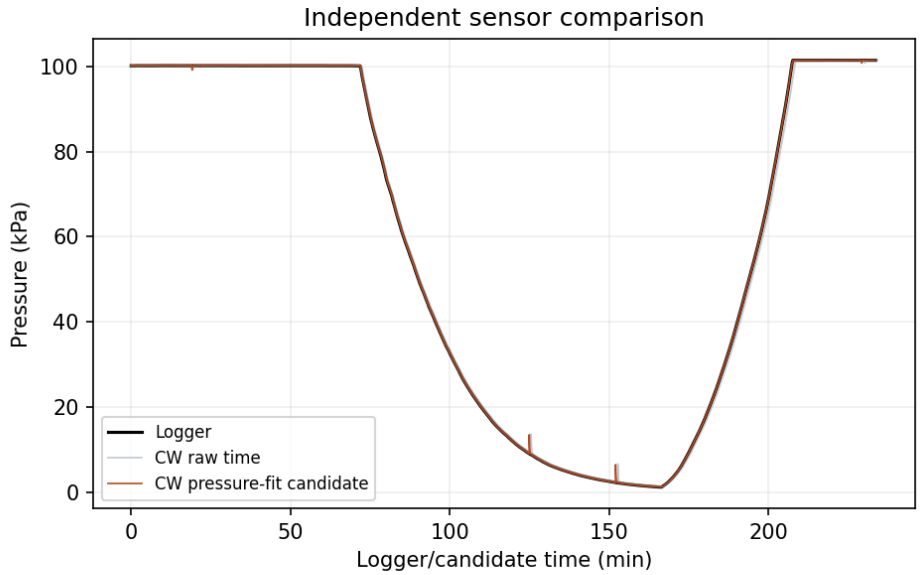


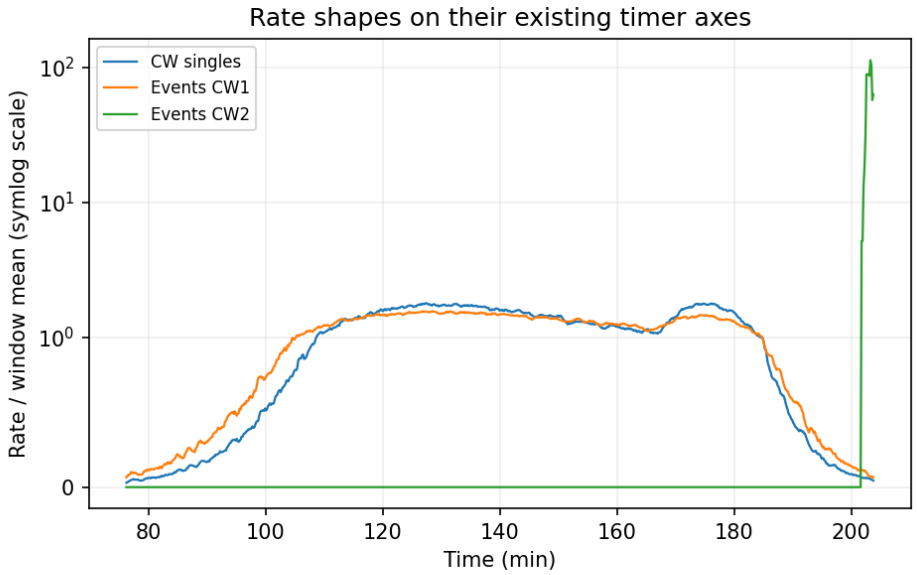


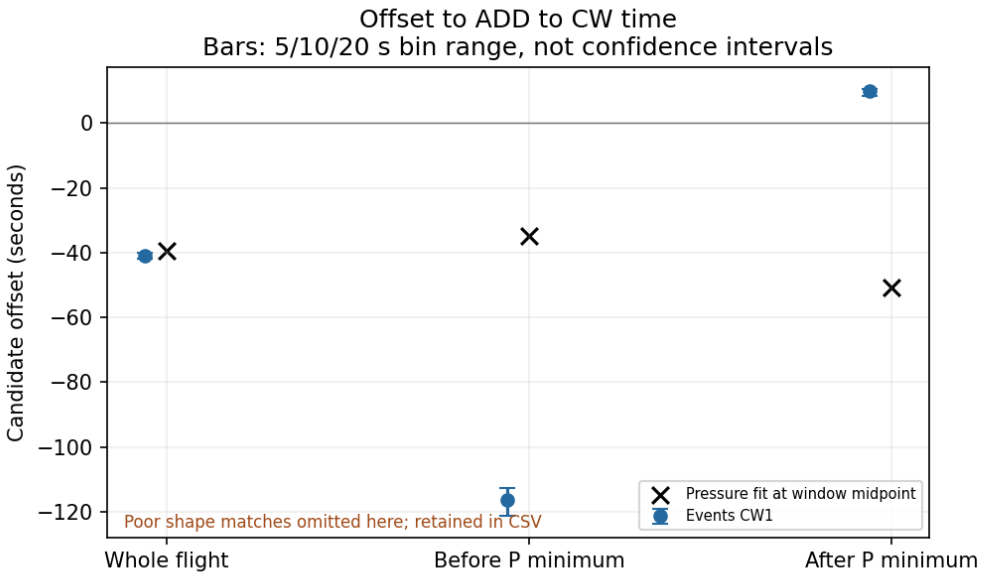


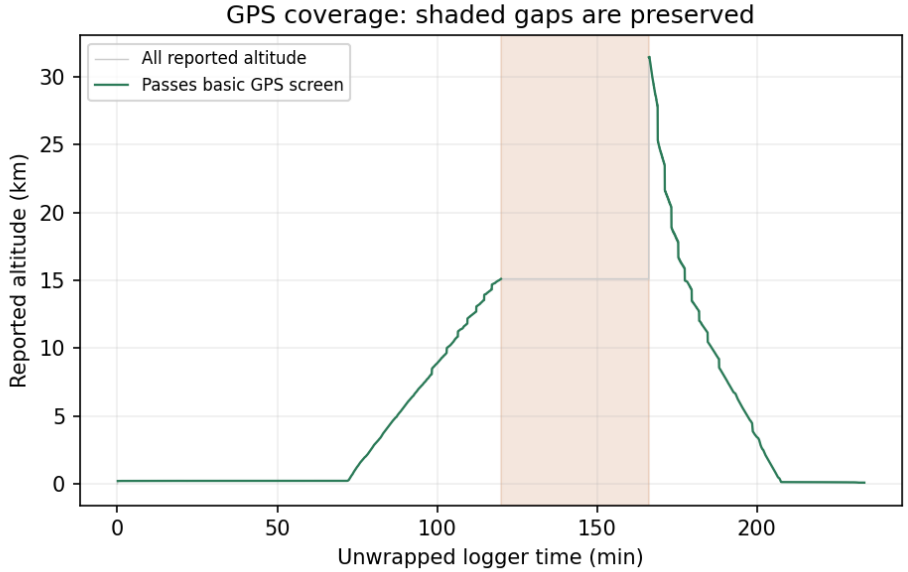
June: count-shape offsets depend on the logger channel and flight portion. Pressure estimates are an independent diagnostic, not verified timing.

2025-10-04 | Timing comparison — diagnostic candidates only



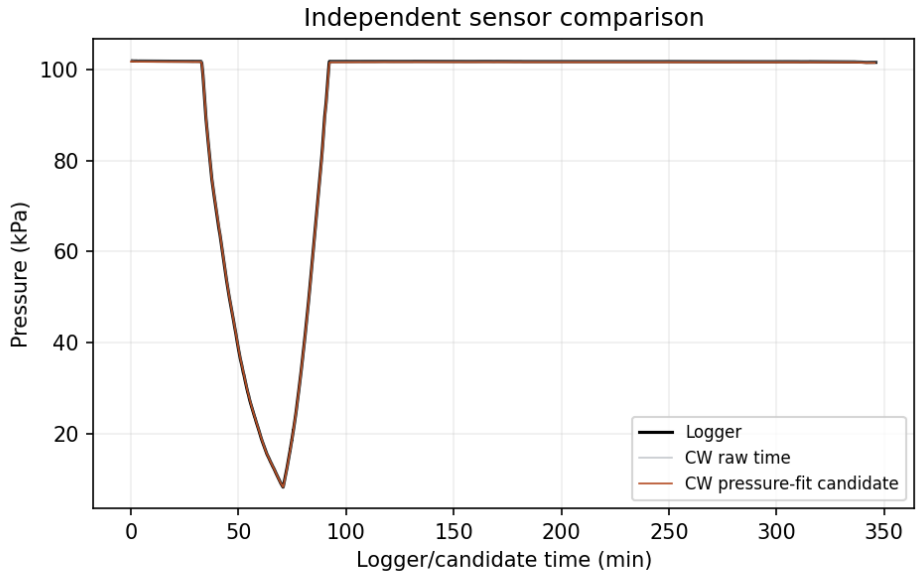


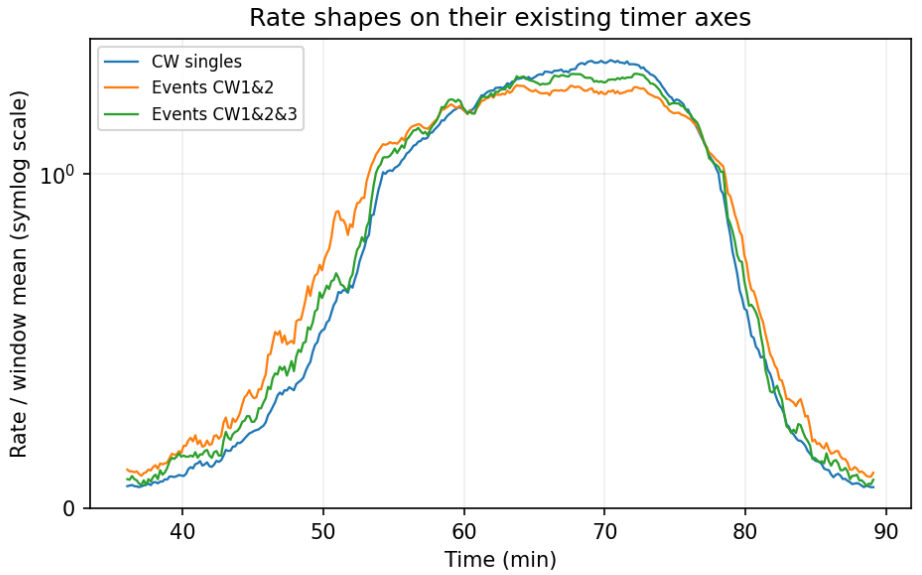


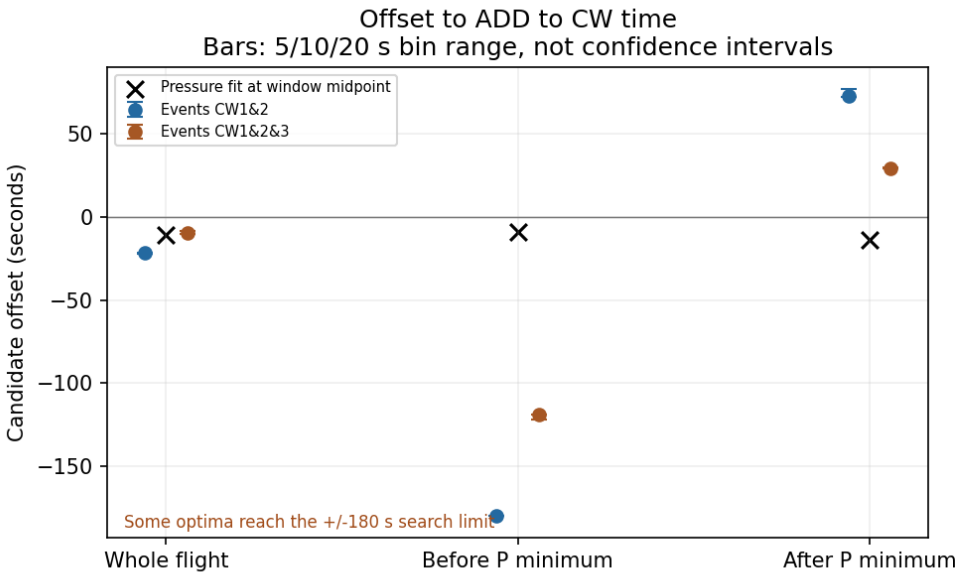


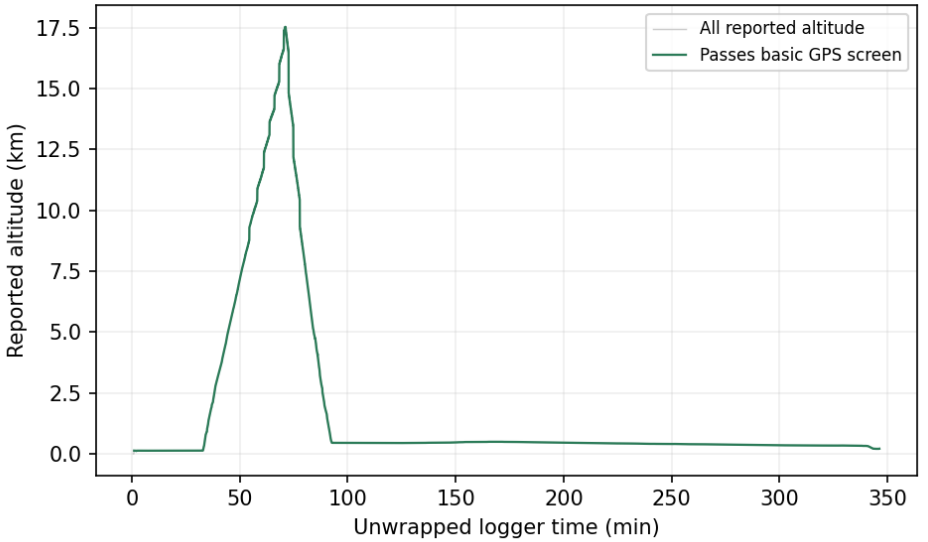
October: CW2 has a distinct late burst and fails the shape-match screen. The onboard GPS coverage gap remains visible and is not interpolated.

2026-03-28 | Timing comparison — diagnostic candidates only









March: offsets differ across the two coincidence channels and flight portions; some searches reach the +/-180-second limit. No single verified correction is established.